

Exercise 8.2

Inheritance of One Pair of Characteristics (Incomplete Dominance)

INTRODUCTION:

When a pure breeding, red coated shorthorn bull is crossed with a pure breeding, white coated cow, the offspring always has a roan coat. Neither coat colour is completely dominant so that both colours are expressed in the hybrid offspring. This is an example of *incomplete dominance*.

Procedure:

In this exercise you will use the punnett square method to show diagrammatically the inheritance of one pair of characteristics where one allele is incompletely dominant to the other.

Study the type example carefully then work through some examples yourself.

Type example: In shorthorn cattle, red coat colour is incompletely dominant to white coat colour. Determine the genotypes and phenotypes of the F_1 and F_2 generations if a pure breeding, red coated shorthorn bull is crossed with a pure breeding, white coated shorthorn cow.

Let the symbol "R" represent the gene for the red coated character.

Let the symbol "W" represent the gene for the white coated character.

*As there is incomplete dominance, both alleles can be represented by capital letters.

Parents: phenotypes red coated bull $\times$ white coated cow

genotypes	RR	WW
Possible gametes	R and R	W and W
1st Generation Offspring (F_1)		W W
	R	RW RW
	R	RW RW

genotypes	all RW
phenotypes	all roan hybrids
Parents: phenotypes	Roan bull $\times$ roan cow
genotypes	RW RW
Possible gametes	R and W R and W
2nd Generation Offspring (F_2)	R W
	R RR RW
	W RW WW

HOMEWORK:

- (1) A pure-bred wire haired terrier is crossed with a pure-bred smooth-coated dog. Given that, in dogs, wire hair is dominant to smooth hair, determine the genotypes and phenotypes of the F_1 and F_2 generations.
- (2) In guinea pigs, rough coat is dominant to smooth coat. Determine the possible genotypes and phenotypes of the offspring if two rough coated hybrid guinea pigs are mated.
- (3) In fruit flies, red eyes are dominant to white eyes. Determine the possible genotypes and phenotypes of the offspring if a pure breeding red eyed fruit fly is crossed with a hybrid red eyed fruit fly.

FOR THE RESEARCH STUDENT:

Given that, in guinea pigs, black coat is dominant to white coat, you can have three possible genotypes for coat colour in guinea pigs e.g. BB; Bb and bb. However, guinea pigs with the genotypes BB and Bb have the same phenotype, black coat colour. Find out what is meant by a "test cross" and show how it can be used to determine if a black coated guinea pig is homozygous (BB) or heterozygous (Bb).

BIOLOGICAL TERMS TO LEARN: Gregor Mendel, dominant, recessive, homozygous, heterozygous, pure breeding, hybrid, genotype, phenotype, punnett square.

RESEARCH REPORT:

genotypes $\frac{1}{4} RR + \frac{1}{2} RW + \frac{1}{4} WW$ or $1 RR : 2 RW : 1 WW$:

phenotypes $\frac{1}{4}$ red coat + $\frac{1}{2}$ roan coat + $\frac{1}{4}$ white coat
or 1 redcoat : 2 roan coat : 1 white coat

TRIAL QUESTION: In Andalusian chickens, the black Andalusian character is incompletely dominant to the white-splashed Andalusian character. The heterozygous condition produces blue Andalusian chickens. Determine the genotypes and phenotypes of the F_1 and F_2 generations if a pure breeding, black Andalusian is crossed with a pure breeding, white-splashed Andalusian.

Given that, in Snapdragons, red flower colour is incompletely dominant to white flower colour (heterozygotes are pink) and that in shorthorn cattle, red coat is incompletely dominant to white coat (heterozygotes are roan) answer the following questions:—

- (1) A pure breeding, red flowered snapdragon is crossed with a pink flowered snapdragon. Determine the possible genotypes and phenotypes of the offspring.
- (2) A roan shorthorn bull is crossed with a roan shorthorn cow. Determine the possible genotypes and phenotypes of their offspring.
- (3) A pure breeding, red coated shorthorn bull is crossed with a roan coated, shorthorn cow.
 - (a) What is the chance (probability) that they will produce a roan calf? Explain.
 - (b) What is the chance that they will produce a white calf? Explain.

HOMEWORK:

- (1) Define the term incomplete dominance
- (2) Try to explain why it would be easier for a geneticist to work with characters exhibiting incomplete dominance than characters exhibiting dominance and recessiveness.
- (3) In four-o'clock plants, if a pure breeding red flowered plant is crossed with a pure breeding white flowered, the offspring have pink flowers.
 - (a) Explain the type of inheritance? Illustrate your answer diagrammatically.

- (b) Determine the genotypes and phenotypes of the offspring if two pink flowered four-o'clock plants are crossed.

BIOLOGICAL TERMS TO LEARN: incomplete dominance.

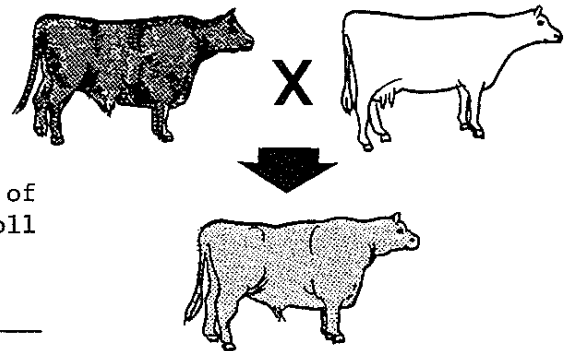
incomplete & co-dominance

Many characteristics studied in genetics are controlled by genes which are dominant or recessive.

However, not all characteristics follow this pattern.

Here is one example:

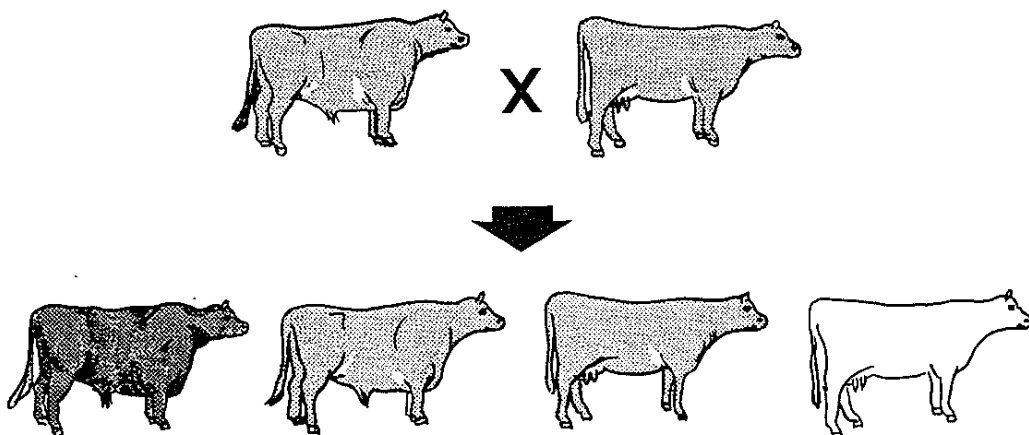
- Describe the offspring from the mating of a white poll shorthorn cow and a red poll shorthorn bull.



- Can you describe either red or white coat colour in poll shorthorn cattle as a dominant feature? _____
- Give a reason for your answer: _____

This case does *not* fit the dominant/recessive model. We describe this as a case of *CO-DOMINANCE*. The *hybrid* offspring shows the effect of both the red and white genes and is called *roan*.

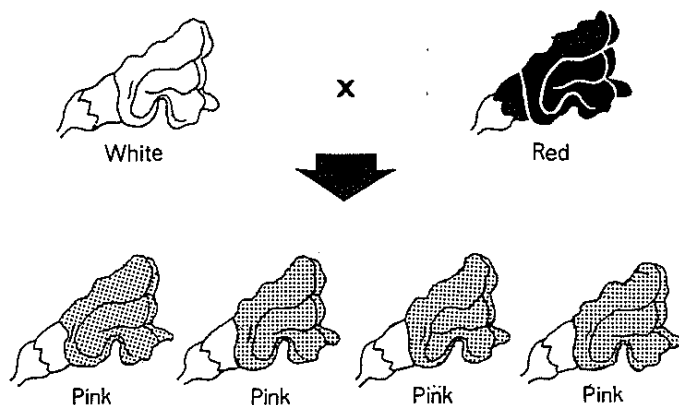
What type of offspring are produced if a *roan* poll shorthorn cow is mated with a *roan* poll shorthorn bull?



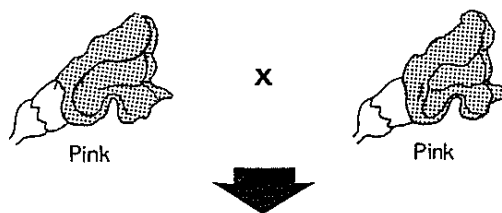
- List the offspring produced by this mating.

(a) _____ (b) _____ (c) _____ (d) _____

When red-flowering snapdragon plants are crossed with white-flowering snapdragon plants, pink-flowered plants result. This is a case of *INCOMPLETE DOMINANCE*. The hybrid is *intermediate* between the parent types.



- In the spaces provided show the result of a pink $\times$ pink snapdragon cross.



(a) _____ (b) _____ (c) _____ (d) _____

- What results would you expect if you were to cross red-flowering plants with pink-flowering plants?

- Why are no white flowers produced?

- Explain why there are *three* possible offspring from crossing 2 *hybrids* in the cases of incomplete and co-dominance.

